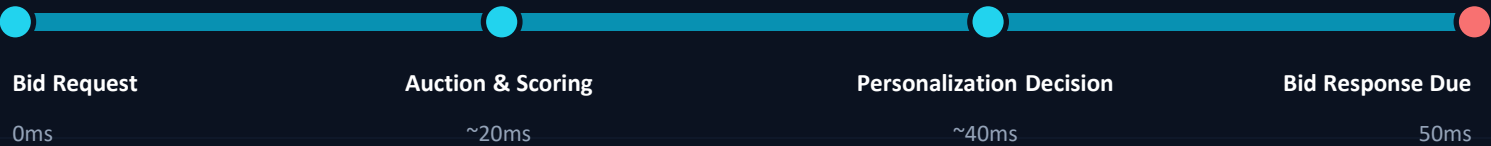


The Personalization Latency Crisis

Real-time bidding exchanges score, decide, and act on a user's intent inside a single round trip — and that round trip is shrinking faster than legacy personalization stacks can follow.

THE RTB BID LIFECYCLE



50ms is the entire budget — and legacy pipelines spend most of it waiting.

WHY MILLISECONDS MATTER



< 50 ms

Full RTB bid lifecycle SLA, end to end



Millions / sec

Behavioral events generated across the platform

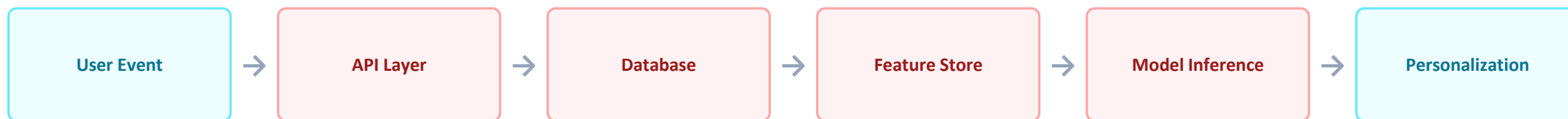


Stale

Profile state the moment latency budget is missed

Every millisecond of avoidable latency is a bid an advertiser cannot win, and a moment of intent a brand cannot act on.

Why Existing Personalization Architectures Break Under RTB Load



4 of 5 stages perform blocking, synchronous I/O — each one adds latency the bid cannot afford to spend.



Blocking Operations

Sequential DB round trips and synchronous calls stall the request thread on every lookup.



High Infrastructure Cost

Thread-per-request models and always-on inference scale compute cost faster than traffic.



Latency Accumulation

Each pipeline stage adds tail latency; P99 degrades sharply long before P50 looks unhealthy.



Poor Scalability

Lock contention and connection pool exhaustion turn traffic spikes into cascading failures.

10–50ms+ Typical legacy pipeline P50 latency

Linear Cost scaling with concurrent traffic

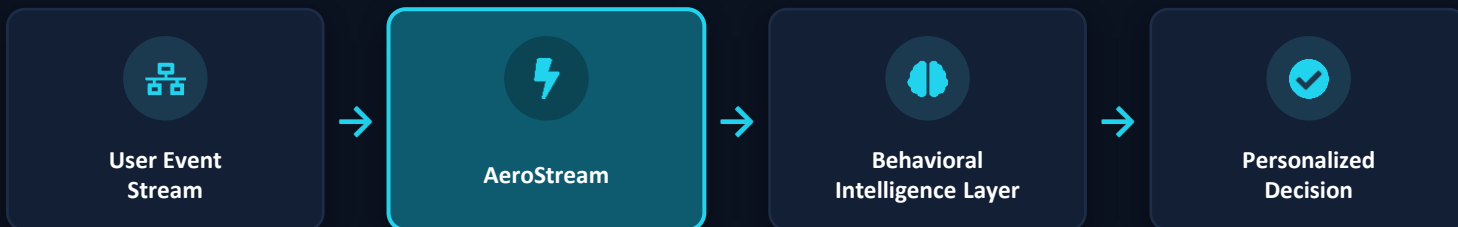
High Lock contention under thread-per-request

PRODUCT INTRODUCTION

Introducing AeroStream

Real-Time Behavioral Intelligence for Hyper-Personalization at Scale

AeroStream separates event ingestion from behavioral computation — accepting every event in microseconds while a speculative intelligence layer resolves and enriches user profiles in the background, never blocking the bid path.



KEY INNOVATION



Decoupled Ingestion

Events are accepted and validated independently of profile scoring.



Speculative Intelligence

Background workers predict and pre-compute before the next request arrives.



Sub-Millisecond Path

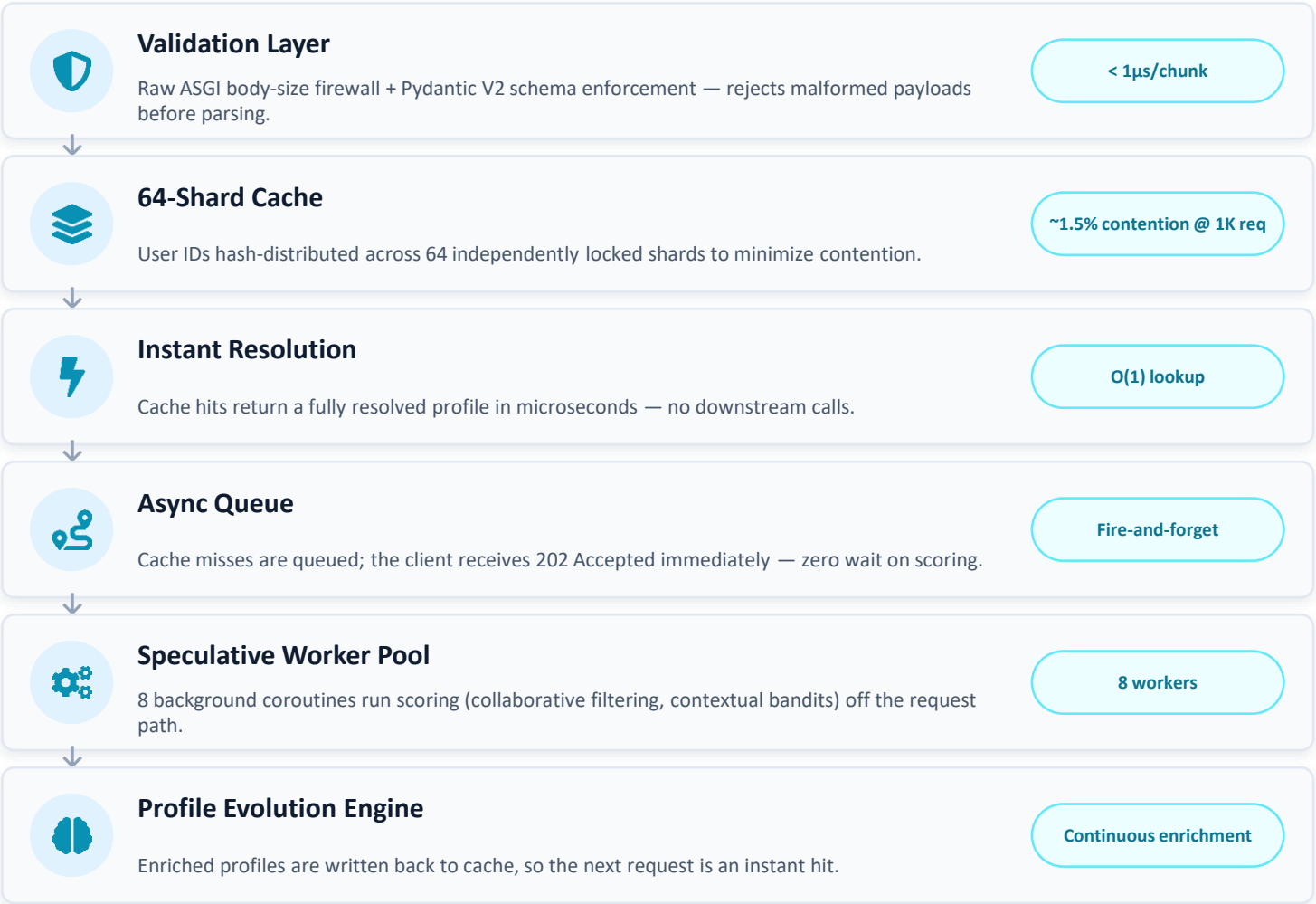
Cache-resolved requests return in microseconds, not milliseconds.

Value Proposition *Personalization that keeps pace with the bid — not the other way around.*

Built on FastAPI, Uvicorn, and asyncio — engineered for the AdTech traffic patterns Epsilon already runs.

How AeroStream Works

A non-blocking pipeline from wire bytes to behavioral insight



WHY THIS DESIGN WINS

- ⚡ Async-first, not thread-per-request**
FastAPI + Uvicorn + httptools run on a single cooperative event loop — thousands of connections, zero thread contention.
- 📦 Sharded, not global, locking**
64 independent asyncio.Lock shards mean concurrent requests almost never wait on each other.
- ⚙️ Speculative, not reactive, scoring**
Profile enrichment happens before it's needed, so the hot path almost always hits a warm cache.

Result: request path latency is bounded by cache lookup time, not by model inference or database round trips.

Proof of Performance

Live server, dirty-fuzzer and concurrent stress conditions — sub-1ms SLA target



66µs
P50 LATENCY



610µs
P99 LATENCY



8,921
EVENTS / SECOND



73.8%
CACHE HIT RATE

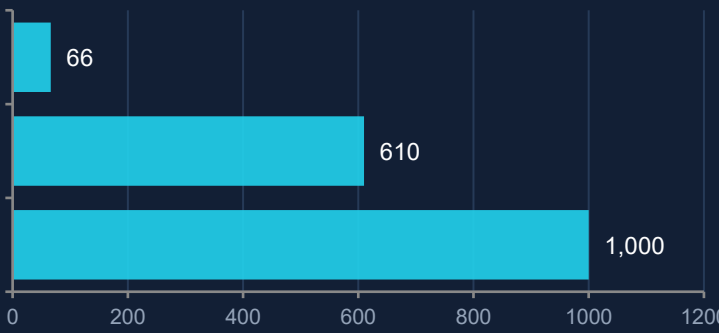


2.14×
SPEEDUP RATIO



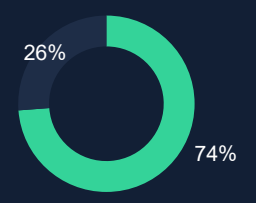
100%
FUZZER BLOCKS (52/52)

LATENCY vs. SLA TARGET



Metric	Value
P50 Achieved (66µs)	66
P99 Achieved (610µs)	610
Target SLA (1,000µs)	1,000

CACHE EFFICIENCY



3,690 / 5,000 requests served from memory

- Served from Cache
- Cache Miss



100%
ADVERSARIAL SUCCESS RATE

52 of 52 fuzzer probes blocked — schema poisoning, payload bombs, and boundary attacks all rejected at the perimeter.

482.45ms of cumulative processing time saved via caching across the benchmark run.

Business Impact for Epsilon

From engineering benchmark to platform-level economics



Operational Efficiency

A single async event loop replaces thread-per-request handling — fewer moving parts, fewer on-call pages.



Infrastructure Reduction

Speculative caching cuts redundant downstream calls; 73.8% of requests never touch the inference path at all.



Faster Personalization

66μs P50 leaves over 49ms of the RTB budget for auction logic — instead of consuming it on lookups.



Lower Cloud Costs

Cache-first resolution means fewer GPU/inference calls per impression at the same traffic volume.



Better User Experiences

Profiles stay fresh via continuous background enrichment, not stale snapshots recomputed on a delay.



Future AI Readiness

The speculative worker pool is a natural seat for larger models — swap in heavier scoring without touching the hot path.

Bottom line: AeroStream turns a hard latency constraint into a competitive margin — faster decisions, fewer compute cycles, fresher profiles.

The Future of Hyper-Personalization

AeroStream is positioned as foundational infrastructure, not a point solution.

01



Real-Time Profile Resolution

Sub-millisecond cache-backed identity resolution at production scale.

SHIPPED

02



Behavior Prediction

Speculative workers evolve from rule-based scoring to learned prediction models.

NEXT

03



Cross-Channel Customer Memory

Unify behavioral profiles across web, app, and offline touchpoints.

PLANNED

04



Enterprise Scale Deployment

Multi-region sharding and horizontal worker scaling for Epsilon's full traffic volume.

PLANNED

AeroStream transforms personalization from reactive targeting into predictive engagement.